State coupling in reduced syn3A — Core A′, with tRNA charging in the ODE block

fig 1r redrawn with the lumped charging step where the scoping note places it · 22 metabolic reactions · 17 genes · every other node sits where fig 1 puts it

CME block

Gillespie / Lattice Microbes · stochastic · not differentiable

three modules: 17 × 3 reactions plus one translocation

ODE block

odecell → LSODA · deterministic · **190 → 22 reactions**

four modules: three of six survive, plus tRNA charging moved in from the CME; ~32 dynamic states and 5 chemostats

the Core A′ reduction	
✗	**removed** · the module is cut, and every edge touching it is greyed
	changed · the module survives in reduced form; the box carries **old → new**
———	**greyed edge** · the coupling is gone. Everything still coloured is live in Core A′
▭	**moved** · one module crosses from the CME to the ODE block; its box is the only one not at fig 1's position

DNA replication ✗

DNApol3 = 35, frozen

gene copy number

k_replication
elongation from live dNTP conc.

17 mRNAs

mRNA decay
455 → 17 mRNAs

atp, datp, dctp, dgtp → 1

RnaPconc = 187, frozen

transcription
455 → 17 genes

amp, atp, cmp, gmp → 1

atp, ctp, gtp, utp

20 uncharged tRNAs

17 mRNAs

DnaA
initiates replication

atp, gtp · charged tRNA

ribosomeConc = 503, frozen

translation
455 → 17 genes

k_transcription
per gene, from live NTP conc.

k_translation
per gene, from the live charged-tRNA pool

1 membrane protein locus (ptsG)

the hook — state
every **1.0 s** of cell time

counts → ODE initial conditions (partTomM)
integrate 1 s (LSODA)
write counts back (mMtoPart)

~6,300 calls per cell cycle
gradients do not cross here

debit NTP pools
clamped at zero, deficit carried
a max(0,·) on the interface

credit NMP
mRNA decay recycling,
unclamped

protein counts → enzyme conc.
GPR: 87 single-gene · 4 OR (sum) · 5 AND (min)
all six ODE modules; drawn once

the catalytic channel is not cut. Fig 1 draws it once, into Cofactor, which Core A′ deletes — hence the grey edge above. In Core A′ it runs into **Transport, Central and Nucleotide**: 17 single-gene GPRs, one-to-one, no AND/OR logic needed.

datp, dctp, dgtp, dttp

atp, gtp *ctp, utp chemostatted*

V rescales every count↔mM conversion
growth dilutes all four ODE modules; drawn once

fmet-tRNA

the CME rebuild — rate constants
every **60 s**, MinCell_restart.py

live pools → k_transcription, k_translation
the whole CME is reconstructed

k_replication branch gone with replication
still the only ODE→CME direction

CellV → getConc

growth medium
17 → 5 clamped concentrations
glucose_e (40 mM), CTP, UTP, aa pool, O_2

no feedback on the medium

debit ATP
hydrolysis: trsc, translat, mRNAdeg, transloc

Transport
87 rxns → 6 rxns
PTS cascade ×5 · lactate export

g6p, pyr

lac, pep

Central
26 rxns → 10 rxns
glycolysis through lactate · NOX dropped

pyr
ade, adn +10

aa pool chemostatted

3pg, pyr
13dpg, pep

g6p

g6p +1

Nucleotide
46 rxns → 5 rxns
PGK3, PYK3 · ADK1, PPA, GK1

10fthf, glu +4

ctp, utp
dtmp, udp +1

prpp

Lipid ✗
rxns

debit charged tRNA · credit uncharged
~553 residues/s against a pool of order 10^3
a buffer of seconds — check 7's counter to watch

glu, h2o

debit fmet-tRNA
the one aa-tRNA
pulled at the hook

tRNA charging
20 × 5 rxns → 1 lumped step
CME → ODE
owns the charged and uncharged pools
ATP → AMP + PP_i, into recycling's pools
lumping and formalism ours;
placement the scoping note's

Cofactor ✗
rxns

Amino acid ✗
rxns

why charging moved. Fig 1 draws it in the CME because the published model runs it there. The scoping note always placed the lumped step in the ODE block, and the frozen registry marks both tRNA species metabolite-scale; the wave plan mis-assigned it and fig 1r inherited that. Spec §12 amendment 1. Its 553 events/s were 353× every other stochastic event combined.

9 lipid headgroups

charged-tRNA pool

fmettrna

currency pool ATP · ADP · AMP · P_i · PP_i · NAD(P)(H)
not a module — a drawing device. Now shared by **four** modules — Transport, Central, Nucleotide, tRNA charging. NAD(P)(H) survives through LDH_L alone.

growth — surface-area accounting
every 1.0 s, in neither block

~~CellSA_Lip = 0.513 × Σ (lipid count × headgroup area)~~
CellSA_Prot = 28.0 nm² × (membrane protein count)
r = √(CellSA/4π) V = (4/3)πr³

ptsG is the only membrane protein: 831 × 28 nm².
V reaches ~1.07× — the 2× clamp is never approached.
Report fractional growth, never a doubling time.

V clamped at 6.70e-17 L = 2× initial.
There is no division event.

edge kinds — what actually crosses each interface	
———	**mass** · shared state, continuous; gradients cross inside the ODE block
- - -	**currency** · routed via the pool node
- - -	**deferred counter** · CME accrues a cost, the hook debits it one step later
· · · ·	**catalytic** · counts enter rate laws as parameters, no mass flows
- - -	**rate constant** · pools re-enter the CME at the 60 s rebuild, piecewise-constant
———	**volume** · counts set surface area, which sets V, which rescales concentrations
- - ⊣	**clamped** · a real dependence replaced by a constant